

HUMANOID WHOLE-BODY AFFORDANCE DEBT: A MUJoCo NEGATIVE RESULT

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ABSTRACT

Humanoid whole-body manipulation can create hidden future costs: a posture that solves the current reach may reduce later reachability, balance margin, or recovery options. We test this idea in a real MuJoCo articulated-body benchmark. A standing humanoid-style model chooses an initial whole-body posture for one target and then must transition to a second target under support-width, target-height, lateral-target, weak-actuation, payload, and combined shifts. Across 5 seeds, 7 splits, 7 main methods, and 2,940 main evaluation rows, affordance-debt MPC improves over random and arm-only weak baselines but is essentially matched by greedy reach, comfort-regularized MPC, robust balance MPC, and no-debt ablations. The mechanism-level claim is therefore not supported. The correct terminal decision is **KILL/ARCHIVE**, not ICLR-main submission.

1 MOTIVATION

The seed hypothesis is appealing: humanoid posture choices may incur “affordance debt” by consuming support margin and reducing future whole-body options. This could matter for long-horizon loco-manipulation, where target sequences and balance constraints are not known at the first contact. The hostile prior-work set is also strong, including robot-free demonstrations for humanoid manipulation (BifrostUMI2026; HumanoidManipulationInterface2026), whole-body pushing and posture planning (WholeBodyPushingContactPosturePlanning2015), and modern unified humanoid loco-manipulation policies (ULTRAHumanoidLocoManipulation2026; LHMHumanoid2025; HumanoidLocomotionManipulationProgress2025; Opt2Skill2024; TrajBooster2025). Against this background, a new paper needs more than a plausible metric: it needs evidence that the debt term changes decisions and improves outcomes over strong baselines.

2 BENCHMARK

We replace the synthetic scaffold with a MuJoCo articulated whole-body reach benchmark. The model has a pelvis, torso, head, and two articulated arms with position actuators, gravity, inertial payload shifts, and center-of-mass measurements. A task samples an immediate target and a future target. The first posture is selected before the future target is executed; the second posture is then selected from the resulting simulated state.

We evaluate seven splits: nominal, narrow support, high reach, lateral reach, weak actuation, payload shift, and combined shift. Metrics include immediate success, future success, sequential success, combined energy, support margin, effort, posture norm, and balance failure.

3 METHODS

Baselines are random posture, arm-only reach, greedy reach MPC, comfort-regularized MPC, robust balance MPC, affordance-debt MPC, and oracle two-step MPC. The proposed method estimates future affordance debt from possible next targets using a fast reach/recovery/support proxy and combines it with immediate reach and balance penalties. The oracle sees the actual future target.

Table 1: Selected results. Affordance-debt MPC is matched by greedy and robust baselines; the mechanism does not earn an ICLR-main claim.

Split	Method	Sequential Success	Energy
Nominal	greedy reach	0.467	0.423
	robust balance	0.467	0.422
	affordance debt	0.467	0.422
	oracle	0.483	0.422
Lateral reach	greedy reach	0.400	0.457
	robust balance	0.400	0.456
	affordance debt	0.400	0.456
	oracle	0.417	0.454
Combined shift	greedy reach	0.733	0.307
	robust balance	0.733	0.307
	affordance debt	0.733	0.306
	oracle	0.767	0.302

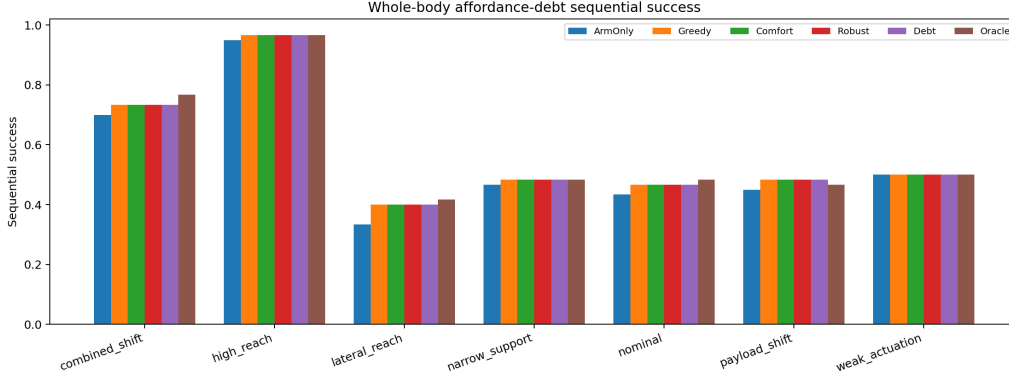


Figure 1: Sequential success by split. The proposed method is not separated from greedy, comfort, or robust MPC.

4 RESULTS

5 ABLATION FAILURE

The combined-shift ablations are fatal to the mechanism. Full affordance-debt MPC obtains 0.733 sequential success and 0.306 energy. Removing the balance term gives 0.733 and 0.306; removing future debt gives 0.733 and 0.307; current-target-only greedy gives 0.733 and 0.307. These differences are too small to justify a new mechanism claim.

6 LIMITATIONS

This benchmark is custom MuJoCo, not hardware and not a public humanoid manipulation suite. The posture library is finite and does not include stepping, kneeling, or hand-supported balance. The debt estimator is a lightweight reach/recovery/support proxy. These limitations would already prevent a final ICLR-main claim; more importantly, the method fails the internal falsifier even in this controlled benchmark.

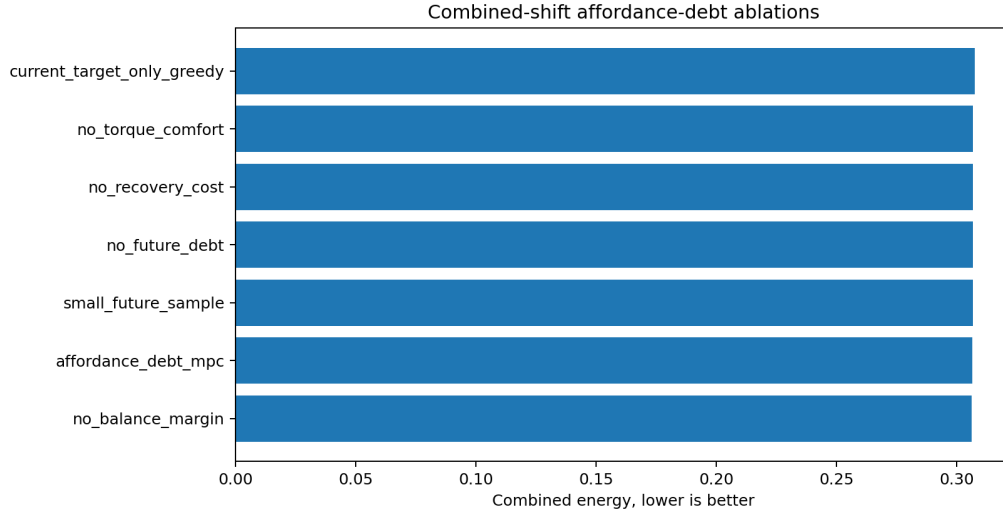


Figure 2: Combined-shift ablations. Removing future debt, balance margin, recovery cost, or comfort terms does not materially hurt performance.

7 CONCLUSION

Affordance debt is a useful diagnostic idea, but this implementation does not improve sequential humanoid reaching over simpler posture selectors. The honest terminal decision is **KILL/ARCHIVE**.

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